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github.com/d3banjan
d3banjan.github.io/engineering

SKILLS

Agent / Eval

Arize Phoenix · butterflow ·
LangChain · LlamaIndex · RAG · vector
DBs · token economics

Distributed

Dask · Celery · Django · PostgreSQL ·
S3/MinIO · Docker · Kubernetes

ML / Research

PyTorch · HuggingFace · LoRA/PEFT ·
GPTQ · CUDA

EDUCATION

B.S.–M.S. Physics

IISER Kolkata, 2007–2012

Doctoral research (not completed)

TU Clausthal, 2012–2016

LANGUAGES

English (fluent) · German (B1)
Hindi (native) · Bengali (native)

Debanjan Basu

SENIOR ML ENGINEER · DISTRIBUTED SYSTEMS · LLM
OBSERVABILITY

Six years production ML engineering at Nexern (Berlin): LLM agent observability (Arize Phoenix), distributed pipelines (Dask/Celery, **5–10×** speedups), GenAI agent deployment. Built **butterflow** — declarative agent eval framework with token caching between runs; **falcon** gates unsafe ML deserialization at the type-checker level with formally verified typestubs.

SELECTED PROJECTS

butterflow

github.com/d3banjan/butterflow

- CLI framework: declarative agent flow definitions, token caching between test runs, evals and cost optimization unified
- Same execution trace, cache aggressively, measure quality simultaneously — user-flow testing and token cost as one problem

falcon

github.com/d3banjan/falcon

- Python typestubs tracking payload annotations by source to gate unsafe ML deserialization (pickle, HDF5) at the type-checker
- Born from huntr.com CVE research on serialization-route vulnerabilities in GenAI platforms; machine-checked soundness proof included

EXPERIENCE

Nexern GmbH

Senior ML & Data Engineer · Aug 2020–Jun 2026 · Berlin

- Built in-house HR platform in **3 months** — multi-entity, multi-jurisdiction across **5 EU/EMEA countries** Django/PostgreSQL, ADR + spec-driven sprints, invariants enforced across backend, frontend and DB
- Built LLM agent observability with Arize Phoenix; trajectory analysis for debugging agent failures in production
- Achieved **5–10×** pipeline speedups via Dask distributed processing, Celery task queues, and vectorised operations
- Led greenfield GenAI agent projects: LLM-based prediction pipelines and automated web data extraction
- Django data platform: large-scale JSON ingestion, web crawlers, Docker deployment, GitLab CI/CD

KUGU Home GmbH

Data Scientist · May 2018–Jul 2020

- Physics-informed heat models and time series forecasting with TensorFlow/XGBoost
- IoT pipeline for hundreds of devices — online changepoint detection, OpenStack/Ansible deployment

TU Clausthal

Doctoral Researcher · 2012–2016

- Extended MD codebase (Fortran/C); published in *Physica Status Solidi A* (DOI: 10.1002/pssa.201532488)

RESEARCH

Independent, Oct 2025–present. **5 preprints** across MoE quantization, LoRA-DPO alignment geometry, RoPE provenance, symmetry survey, and compression methodology. **74 machine-checked theorems** (Mathlib).
d3banjan.github.io/research